



Bricks

A LEGO® BRICK MODEL OF

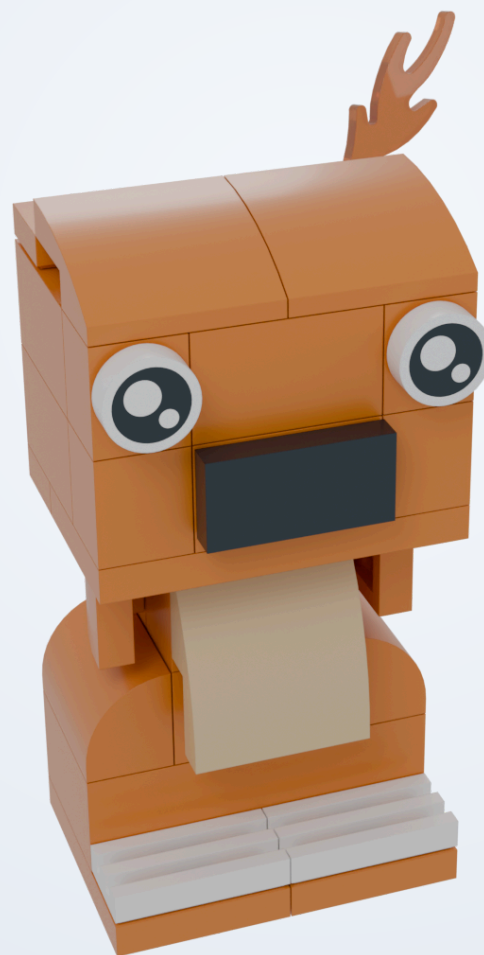
Charmander Starter

A small fire starter with a very big personality.

37 pieces · 15 steps · 1 sections

Unofficial fan-made building instructions

Not physically build-tested. Every part is a real LEGO element and the model was checked on a computer (connections, collisions, every step, balance), but nobody has built it with real bricks yet.



Before you start

An unofficial fan-made manual

This booklet is not a LEGO product. It was generated from the model file by a computer program and checked part by part.

Not physically build-tested

The model and these steps were checked on a computer only: every part/colour is a real LEGO element, all 80 connections between parts were matched in 3D, no two parts overlap, every part can be pushed into place in its step, and the model's balance was estimated. Not tested: how firmly parts hold, part tolerances and how it copes with handling. Take your time and support the model while you work.

The finished model

About 6.1×6.8 cm and 6.5 cm tall, about 28 g (estimated).

Build up from broad clawed feet to the rounded belly and oversized head. Add the little reaching arms, then the curling tail and its real LEGO flame. This is an original brick interpretation of the Pokémon character.

You will need

- The 37 pieces in 5 colours listed at the back, with their LEGO element IDs and BrickLink numbers.
- A brick separator for pieces that sit too deep for a finger.

Tips

- Sort the pieces by colour before you start.
- Build the sections in order: each sub-assembly is finished before it is used.
- New pieces in a step have a yellow outline; earlier pieces are drawn paler.

The 5 colours

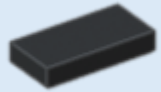
 Orange: 28
 White: 4
 Trans-Orange: 3

 Tan: 1
 Black: 1

How to read the steps

Parts and step number

7



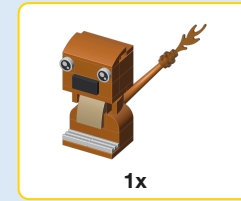
2x

The parts for the step and how many of each, then the step number. The parts of a step are drawn at one scale, so a longer one looks longer; a tile or a long plate also has its size in studs under it.

New parts

The parts added in a step have a yellow outline and are drawn in full colour; the parts from earlier steps are drawn paler.

Sub-assemblies



A yellow-framed picture means: put in the sub-assembly you built in its own section. "Build 4x" means you need four of them.

Sections

Each section starts with its name at the top of the page. Sections are built in the order they appear.

Seen from below

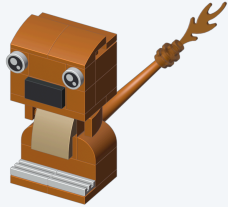
When a step adds parts underneath, the picture shows the model from below. A small label says so (and "seen from behind" likewise). A piece the picture cannot show gets a small picture beside it, ringed in yellow, from the side it can be seen from: where it clicks in.

Insert direction

Some parts slide in sideways (axles through bearings, gears onto axles). The picture shows where they end up; push them in along their axle.

Build overview

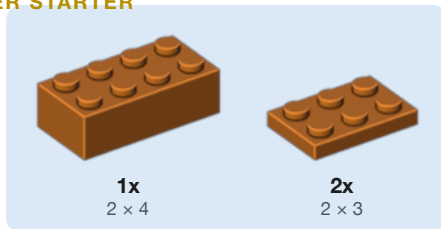
The model is built in 1 sections. Each is finished before it is used.



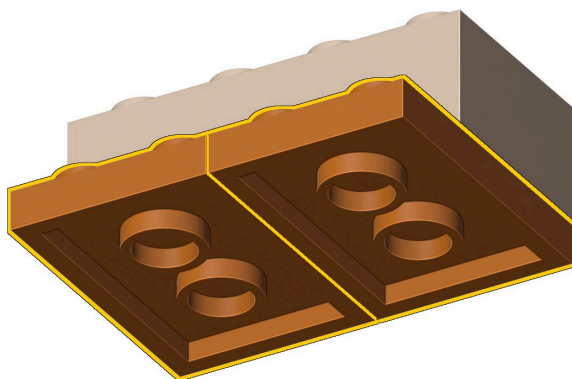
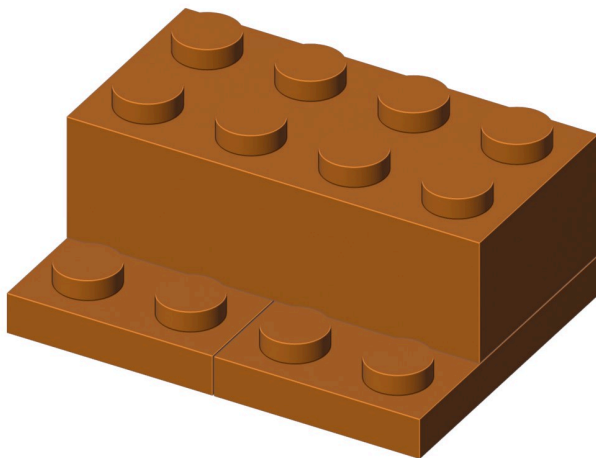
1. Charmander Starter
steps 1–15

CHARMANDER STARTER

1

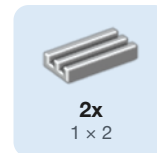


Put the two orange feet side by side and join them with the orange hips

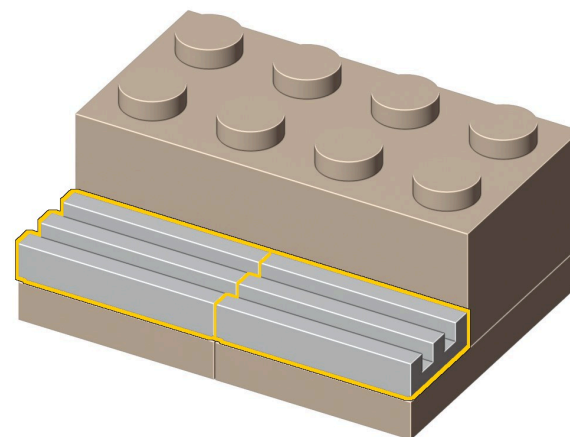
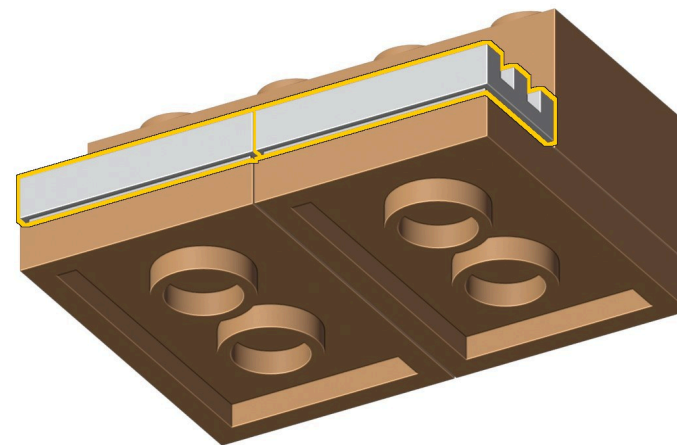


From below: the outlined pieces click in here.

2



Give each foot three little white claws

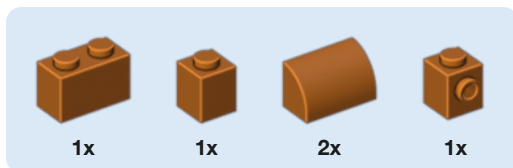


From above: the outlined pieces.

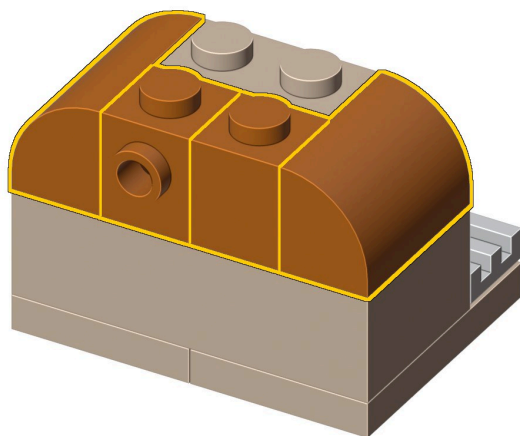
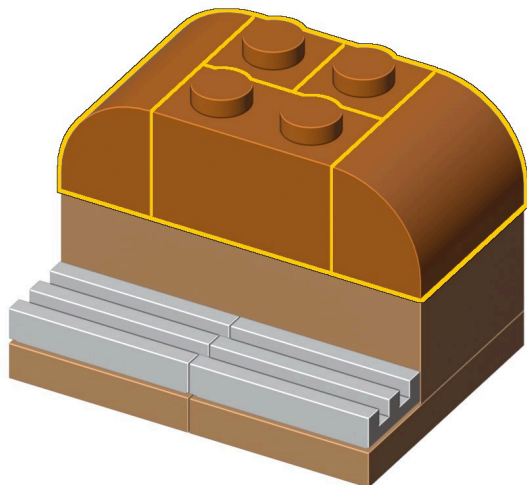
Charmander Starter

seen from below

3



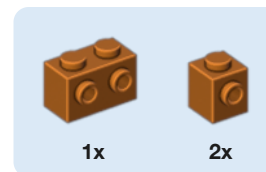
Build the waist and the hollow rear stud for the tail



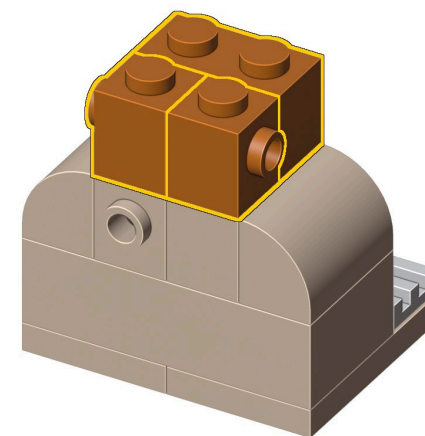
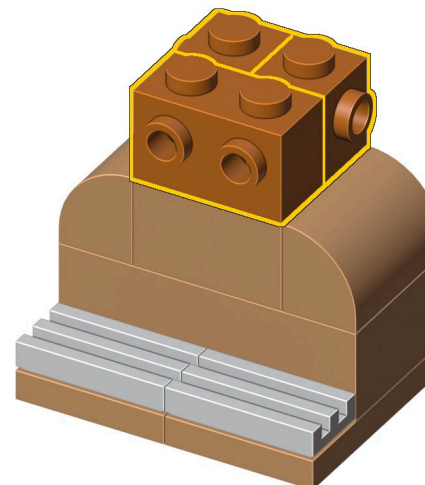
From behind: the outlined pieces.

4

Charmander Starter



Add chest studs in front and shoulder studs at the sides

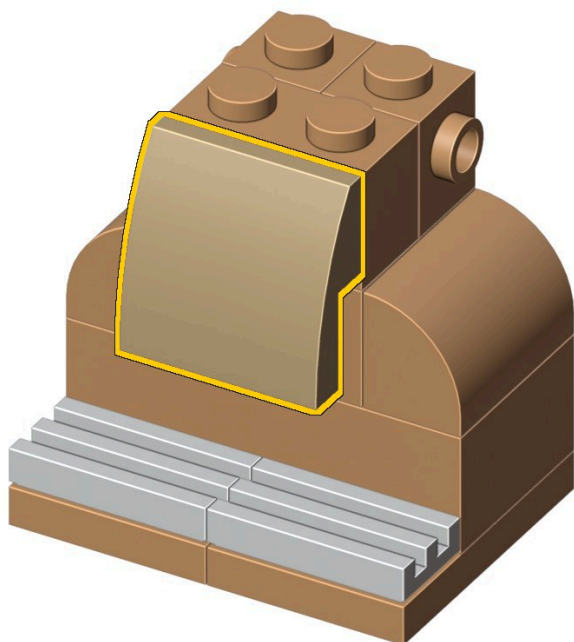


From behind: the outlined pieces.

5



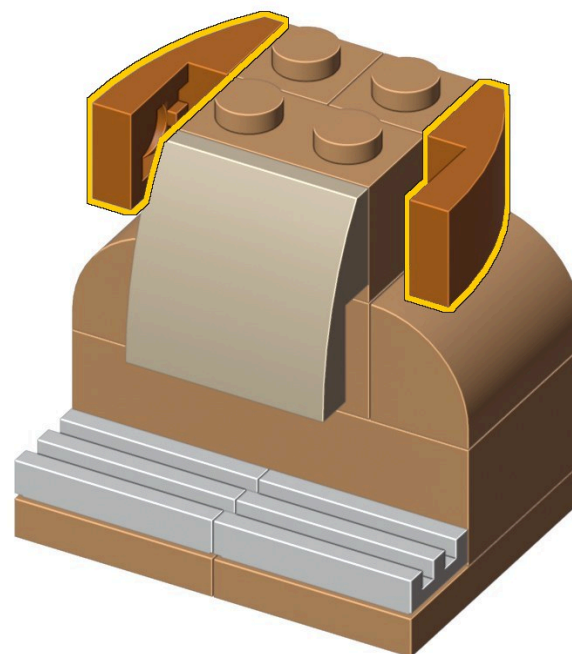
Round off the tan belly



6



Two short arms reach toward a new friend

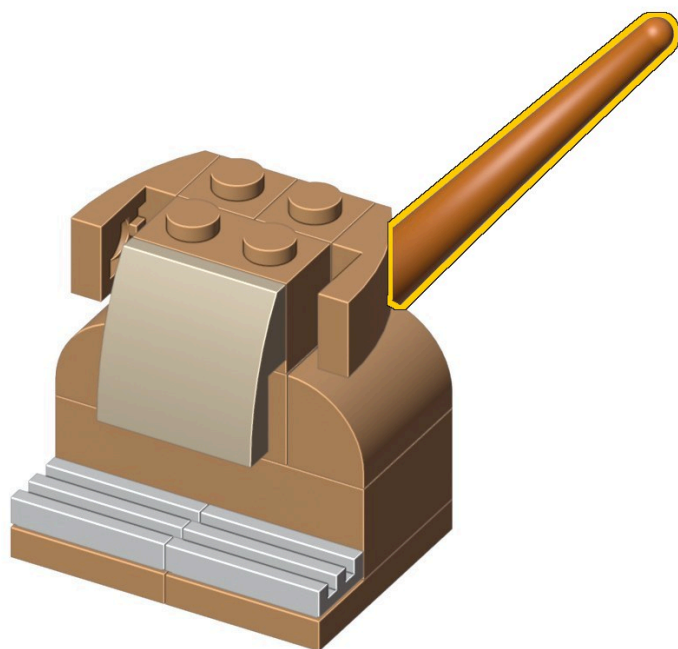


Charmander Starter

7

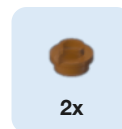


Curl the tail up and out to the side



8

Charmander Starter



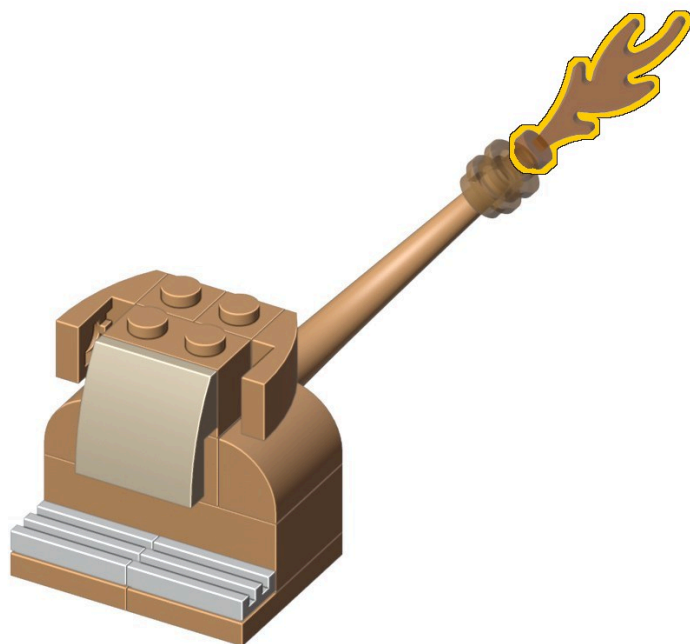
Build a translucent two-plate collar at the tail tip



9

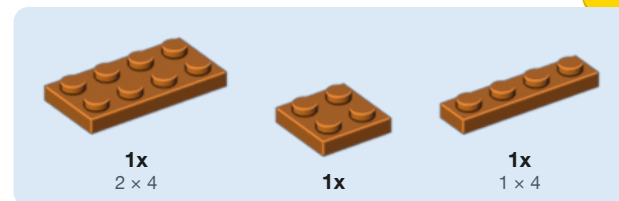


Light the tail with a translucent orange flame

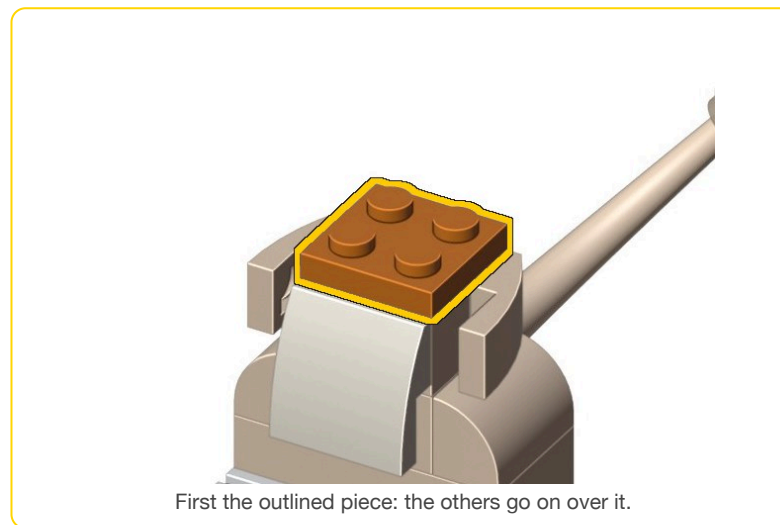


10

Charmander Starter

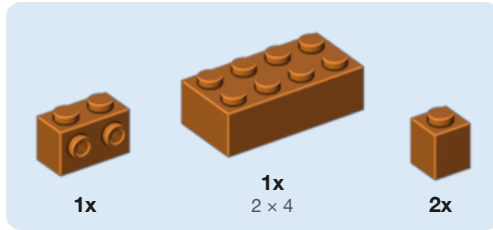


Make a short neck and a broad base for the head

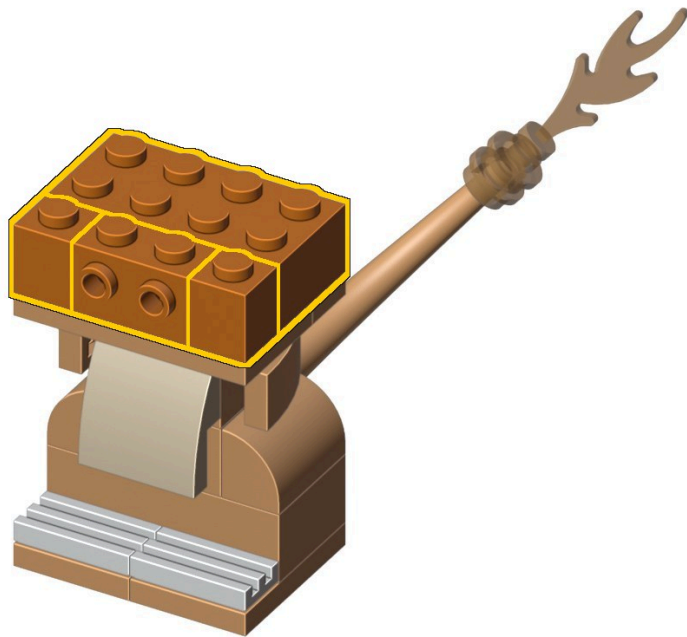


First the outlined piece: the others go on over it.

11

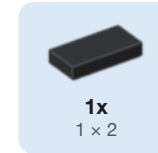


Build the lower jaw and two orange cheeks

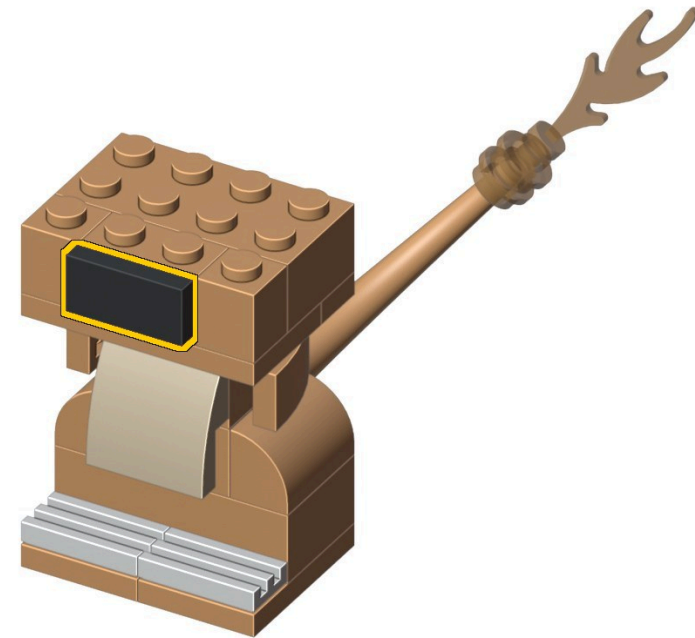


12

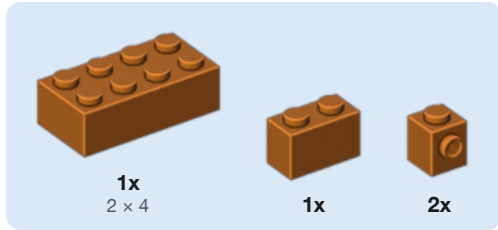
Charmander Starter



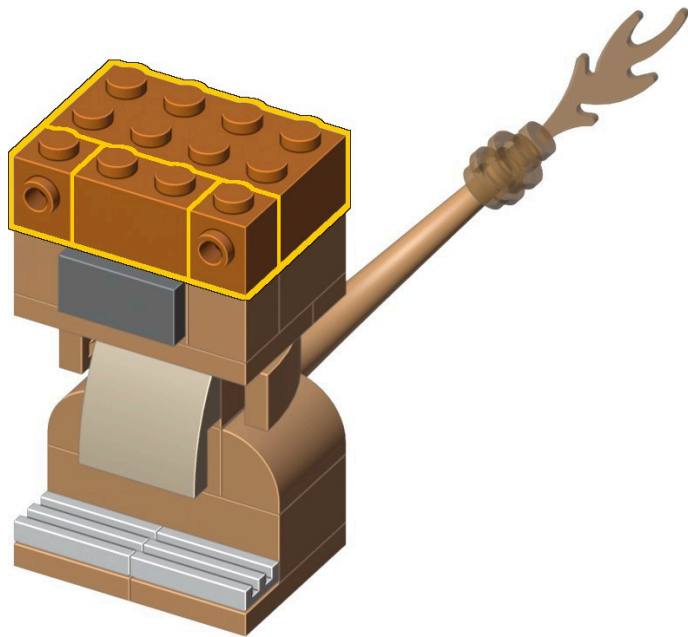
A tiny dark mouth below the eyes



13



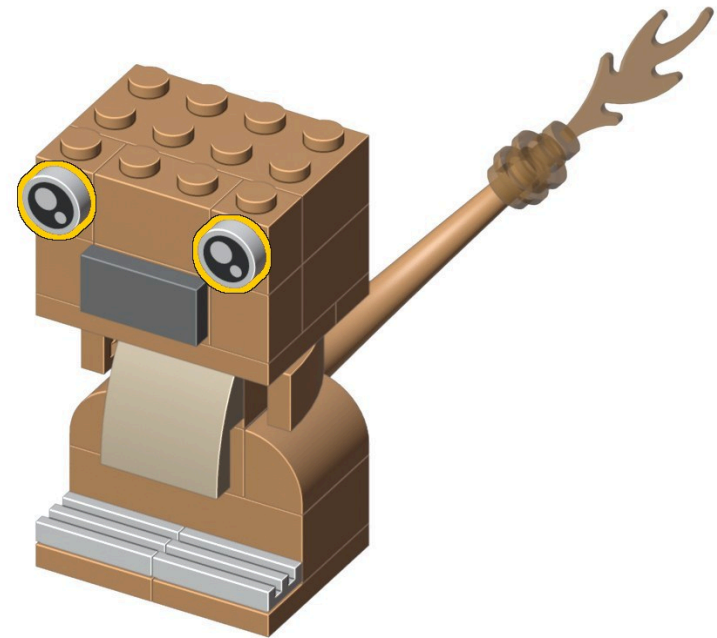
Build the forehead with the eye studs facing forward



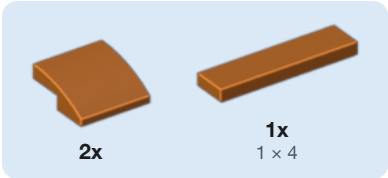
14



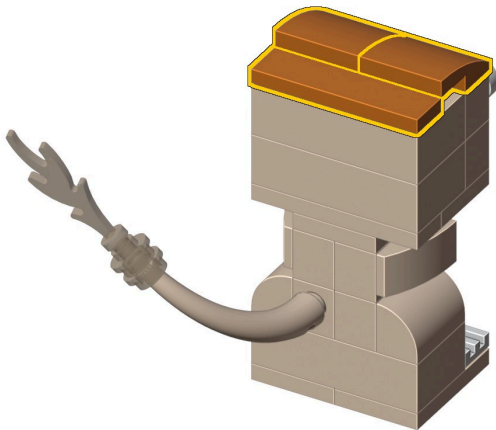
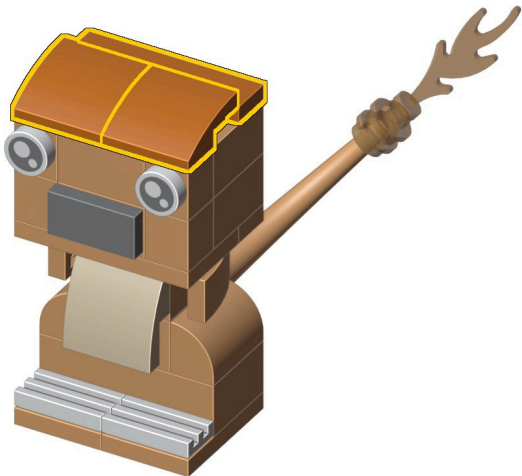
Two bright eyes with their highlights at the top



15



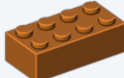
Round off the orange head



From behind: the outlined pieces.

Parts inventory

37 pieces in 20 part/colour lines. Bold: the LEGO element ID (what LEGO Pick a Brick sells); below: the colour and the BrickLink part number.

 1x 306926  Black BrickLink 3069	 1x 6194852  Orange BrickLink 40379	 3x 4173805  Orange BrickLink 3005	 2x 4613981  Orange BrickLink 3004	 3x 4153827  Orange BrickLink 3001	 2x 6385785  Orange BrickLink 37352	 2x 6055069  Orange BrickLink 11477	 2x 6067913  Orange BrickLink 15068
 5x 6431334  Orange BrickLink 87087	 2x 6223454  Orange BrickLink 11211	 1x 4118782  Orange BrickLink 3710	 1x 4613982  Orange BrickLink 3022	 2x 4125278  Orange BrickLink 3021	 1x 4158355  Orange BrickLink 3020	 1x 4160593  Orange BrickLink 2431	 1x 6046924  Tan BrickLink 15068
 2x 6510162  Trans-Orange BrickLink 85861	 1x 6167741  Trans-Orange BrickLink 6126b	 2x 6431716  White BrickLink 98138pb367	 2x 241201  White BrickLink 2412b				

About this model

Build up from broad clawed feet to the rounded belly and oversized head. Add the little reaching arms, then the curling tail and its real LEGO flame. This is an original brick interpretation of the Pokémon character.

Checked on a computer

PASS	real elements	20 part/colour combinations: 0 not real, 0 rare
PASS	connections	80 connections (77 stud, 2 bar, 1 pin); 1 separate piece(s)
PASS	collisions	0 overlapping pair(s) among 37 parts
PASS	buildability	35 insertions tried across 1 submodel(s); 0 problem(s)
PASS	stability	about 28 g; centre of mass inside the base (84% margin); tips at 16.8 deg
PASS	mechanism	no moving parts defined
PASS	electrics	no electrics in this model
PASS	technique	0 note(s)

Not yet built with real bricks

These checks catch most mistakes a designer can make, but not all: clutch power, part tolerances, how stiff a long axle is and how the model survives handling can only be judged with real bricks. If you build it, we would love to hear how it went.

Files

3D model (LDraw, opens in BrickLink Studio, LeoCAD and LDCad), parts lists for BrickLink and LEGO Pick a Brick, and an interactive 3D viewer:

bricks.superfun.games/m/charmander_starter

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